Certification of Imitation Controllers Using Learned Lyapunov Functions

Master thesis by Josua Christoph Lindemann
Date: June 22, 2026, Date of submission: 16. August 2026

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2. Review: Hendrik Alsmeier M.Sc.
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$$x_{k+1} = x_k + T_s f(x_k, u_k). \tag{3.1}$$

$$x_{k+1} = x_k + \frac{T_s}{6}(k_1 + 2k_2 + 2k_3 + k_4), \tag{3.2}$$

where the intermediate slopes k_1 through k_4 are defined as:

$$\begin{aligned} k_1 &= f(x_k, u_k) \\ k_2 &= f\left(x_k + \frac{T_s}{2}k_1, u_k\right) \\ k_3 &= f\left(x_k + \frac{T_s}{2}k_2, u_k\right) \\ k_4 &= f\left(x_k + T_s k_3, u_k\right). \end{aligned} \tag{3.3}$$

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For our experimental setup, two benchmark systems are selected: the double integrator and the cartpole. These systems provide a good balance between simplicity and complexity allowing us to certify the stability of the closed-loop system.

7.1.1 Double Integrator

The double integrator is a simple second order linear system, therefore it serves as the ideal baseline system in the learning and certification pipeline. The continuous-time dynamics are described by the following state-space equation:

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \tag{7.1}$$

where $x(t) = \begin{bmatrix} p(t) & v(t) \end{bmatrix}^\top$ is the state vector containing the position and velocity of the system, respectively, and $u(t)$ denotes the scalar control input (acceleration).

For the *discretization* of the continuous-time system, either Euler's method or the RK4 method with a sampling time of $T_s = 0.1\,\mathrm{s}$ is chosen, depending on certification or simulation purposes. Applying Euler's method (Eq. (3.1)) yields the discrete-time dynamics $x_{k+1} = f_{\text{Euler}}(x_k, u_k)$, whereas the RK4 method (Eq. (3.2)) results in $x_{k+1} = f_{\text{RK4}}(x_k, u_k)$.

The *MPC Formulation* is defined based on Eq. (7.1) with a prediction horizon of $N = 10$ steps.

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